

## L31: Ranking and Selection

### Simulation Without a Black Box

# Outline

The Selection Problem

Why Pairwise t-Tests Are Wrong

Indifference Zone

Rinott Procedure

Rinott Constant and Screening

Budget Trade-offs and Practical Advice

Key Takeaways

## The Selection Problem: $k$ Alternatives, Want the Best

### Problem Statement

$k$  designs with unknown means  $\theta_1, \dots, \theta_k$ .

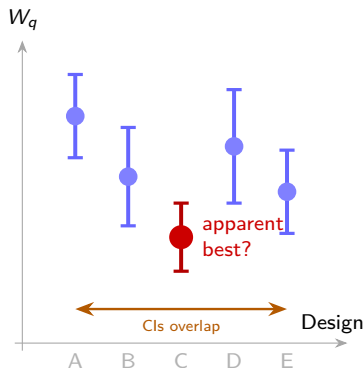
Select the design with the *smallest*  $\theta_i$  with probability  $\geq P^*$  — the **PCS guarantee**.

**Example:** 5 staffing policies; find lowest  $W_q$  with  $P^* = 0.95$ .

- “Best observed”  $\neq$  “statistically best”
- Lowest sample mean may be lowest *by luck*
- R&S procedures control the risk of choosing wrong

No guarantee without R&S

Picking  $\min_i \bar{\theta}_i$  naively has no guaranteed PCS.



# Why Pairwise $t$ -Tests Inflate Error

## FWER Inflation

**Family-wise error rate (FWER):** the probability of making *at least one* false declaration across all comparisons.

With  $k = 5$  designs, there are  $\binom{5}{2} = 10$  pairs. If each test uses  $\alpha = 0.05$ :

$$\text{FWER} \approx 1 - (1 - 0.05)^{10} = 0.40$$

A 40% chance of at least one false declaration!

| $k$ designs | Pairs | FWER (nominal $\alpha = 0.05$ ) |
|-------------|-------|---------------------------------|
| 3           | 3     | 0.14                            |
| 5           | 10    | 0.40                            |
| 10          | 45    | 0.90                            |
| 20          | 190   | $\approx 1.00$                  |

## Warning

Bonferroni correction ( $\alpha/m$  per test) is overly conservative and wastes replications. Use dedicated R&S procedures instead — they are designed for the selection problem specifically.

## The Right Guarantee

R&S procedures give:

$$P(\text{select true best}) \geq P^*$$

*simultaneously* across all  $k$  designs, without inflating the error rate.

# Indifference Zone Formulation

## Indifference Zone Parameter $\delta^*$

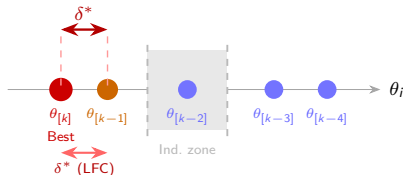
We don't care about differences  $< \delta^*$ . If the best design beats second-best by  $\geq \delta^*$ , we want to select it with probability  $\geq P^*$ .

## Why $\delta^*$ matters:

- Without it, infinite samples needed to separate designs differing by 0.001 seconds
- Set by *domain knowledge*: "We don't care if  $W_q$  differs by less than 1 minute"
- Larger  $\delta^* \Rightarrow$  fewer replications
- Smaller  $\delta^* \Rightarrow$  more replications

## Least Favorable Configuration (LFC)

$\theta_{[k-1]} - \theta_{[k]} = \delta^*$ : second-best and best differ by exactly  $\delta^*$ . R&S guarantees hold at the LFC.



# Rinott's Two-Stage Procedure

## Rinott Procedure

### Stage 1 (pilot):

1. Run  $n_0 \geq 2$  replications for each of the  $k$  designs
2. Compute sample variance  $s_i^2$  for design  $i$

### Stage 2 (second stage):

1. Compute required total replications:

$$N_i = \max\left(n_0, \left\lceil \frac{h^2 s_i^2}{\delta^2} \right\rceil\right)$$

2. Run  $N_i - n_0$  additional reps for design  $i$
3. Select design with best  $\bar{\theta}_i$

$h$  = Rinott constant (from table, depends on  $k$ ,  $n_0$ ,  $P^*$ ).

```
import numpy as np, scipy.stats as st

# Stage 1: pilot
n0 = 20
pilot = {i: run_design(i, n0)
          for i in range(k)}
s2 = {i: np.var(pilot[i], ddof=1)
       for i in range(k)}

# Rinott constant (from table)
# k=5, n0=20, P*=0.95 -> h ~ 2.8
h = 2.8
d_star = 1.0 # 1 minute IZ

# Stage 2: additional runs
N = {i: max(n0,
            int(np.ceil(h**2*s2[i]/d_star**2)))
      for i in range(k)}

all_means = {}
for i in range(k):
    extra = run_design(i, N[i]-n0)
    combined = np.concatenate(
        [pilot[i], extra])
    all_means[i] = combined.mean()

best = min(all_means, key=all_means.get)
```

## Rinott Constant and Kim-Nelson Screening

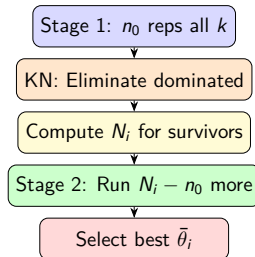
Rinott constant  $h$  at  $P^* = 0.95$ :

| $k$ | $n_0 = 10$ | $n_0 = 20$ | $n_0 = 40$ |
|-----|------------|------------|------------|
| 3   | 2.42       | 2.38       | 2.36       |
| 5   | 2.71       | 2.65       | 2.62       |
| 10  | 2.98       | 2.91       | 2.87       |
| 20  | 3.22       | 3.14       | 3.10       |

Larger  $k \Rightarrow$  larger  $h \Rightarrow$  more replications needed.

**Kim-Nelson screening (KN):** After the pilot stage, *eliminate* designs whose sample mean is provably worse than the current best by more than  $\delta^*$ . Only advance surviving designs to Stage 2.

**Benefit:** Greatly reduces total simulation budget when many designs are clearly inferior.



## Budget Trade-offs: $\delta^*$ and $P^*$

**Trade-off table** (M/M/1 clinic,  $k = 5$ ,  $n_0 = 20$ ):

| $\delta^*$ (min) | $P^*$ | Avg. $N_i$ | Total budget |
|------------------|-------|------------|--------------|
| 0.5              | 0.99  | 312        | 1560 reps    |
| 1.0              | 0.99  | 78         | 390 reps     |
| 1.0              | 0.95  | 52         | 260 reps     |
| 2.0              | 0.95  | 13         | 65 reps      |

Halving  $\delta^*$  *quadruples* the required replications (because  $N_i \propto 1/\delta^{*2}$ ). **Practical advice:**

1. Set  $\delta^*$  from the decision context, *not* from the data (“What difference would change our recommendation?”)
2.  $P^* = 0.90$  or  $0.95$  is usually sufficient;  $P^* = 0.99$  is expensive and rarely necessary
3. Use KN screening to eliminate obviously poor designs early

### Common Mistake

Setting  $\delta^*$  after seeing pilot results — this is circular and invalidates the PCS guarantee. Set  $\delta^*$  before collecting any data.

### Correct Process

1. Consult domain expert: “How small a difference in  $W_q$  would change your staffing decision?”
2. Set  $\delta^* =$  that value.
3. Set  $P^*$  based on risk tolerance.
4. Look up  $h$ ; run the two-stage procedure.



## Key Takeaways

1. **Selection problem:**  $k$  designs, want best with  $P(\text{correct}) \geq P^*$ . Simple best-of- $k$  sample means has no PCS guarantee.
2. **Pairwise  $t$ -tests inflate FWER:** with  $k = 10$ ,  $\text{FWER} \approx 90\%$  at nominal  $\alpha = 0.05$ .
3. **Indifference zone  $\delta^*$ :** set from domain knowledge, not from data; small  $\delta^*$  is expensive ( $N \propto 1/\delta^{*2}$ ).
4. **Rinott procedure:** two-stage; Stage 1 estimates variance, Stage 2 runs  $N_i = \lceil h^2 s_i^2 / \delta^{*2} \rceil$  reps.
5. **KN screening:** eliminate dominated designs after Stage 1 to reduce total simulation budget.
6. **Practical rule:** choose  $\delta^*$  before running anything;  $P^* = 0.95$  is usually sufficient; never set  $\delta^*$  post-hoc.

End of Module M09 — Experimental Design

Next: Module M10 — Verification, Validation, and Advanced Topics.